DataLake/Lakehouse - Architecture Key

"A lot depends not just on how much data you have, of course, but how able you are to make sense of it."

-*Tom O'Rilley*

Unified Platform:

- **Data lakes / lakehouses: raw object storage + curated structured layers**
- **Handle text, events, time series, logs, graph data, and structured tables in one place**
- Hybrid**: schema-on-read - and- schema-on-write** together

- **Rigorous empirical work always depends on coherent, well-governed data!**

Motivation for a Unified Data Platform

Key - reproducibility, data lineage, scalability, and multi-modality

- 70–80% of analytical effort concerns; **locating, reconciling, and shaping** data
- **Schema drift, temporal misalignment, and missingness undermine** reliable inference
- Ensure that **structured, semi-structured, unstructured, graph, and event data** can **not only coexist** but also **interoperate** across the entire ecosystem
- **Meaningful insights** emerge **not from isolated variables**, **but from the relationships** between them
- **Data lakes** preserve **structure, semantics, and temporal relationships as first-class objects.**

Schema - The Blueprint Behind Your Data

Schema-on-Write: <u>Structure First</u>

- **Data is validated, typed, and shaped at ingestion.**
- Guarantees governance, consistency, referential integrity, and data quality
- **Optimized** for **fast updates, indexing, joins, and highly structured queries**

Schema-on-Read: <u>Flexibility First</u>

- Raw data **lands as-is; no predefined schema imposed at ingestion**.
- **Structure is assigned only at query time, depending on the analysis**
- Ideal for semi-structured and unstructured sources:
  - ✓ JSON, events, logs
  - ✓ Text, audio, PDFs, images
  - ✓ Web scrapes, API dumps

Toward Data Lake & Lake house – Some Background

- **OLTP – Operation Databases (Schema-on-write)**
  - **Data must conform *before* being written**
  - **ACID guarantees**, fast transactions, correctness
  - Examples: **PostgreSQL, MySQL, Oracle, SQL Server, MongoDB (schema-ish),** graph DBs, time-series DBs etc.

- **OLAP - Data warehouses (Schema-on-write)**
  - **Denormalized, columnar, still schema-on-write**
  - Data is cleaned, transformed, and modeled *before* loading **(ETL)**
  - Examples: **Snowflake, BigQuery, Redshift**

- **Data lakes / Lakehouses (Schema-on-read & Schema-on-write)**
  - **Raw object storage**
  - Hybrid: **schema-on-read *and* schema-on-write** together
  - Handle text, events, time series, logs, graph data, and structured tables in one place
  - **DataBricks, Delta Lake, Apache Iceberg etc**.

- **Data Lake:** Cheap storage + flexible schema, weak reliability and performance

- **Lakehouse:** Data lake + warehouse-level reliability, transactions, and query performance

Some Specific Challenges Examples

- Financial and macroeconomic series undergo ex-post revisions
- Must version both original and revised states of data
- Temporal versioning enables accurate reconstruction of historical data
- Canonical, documented feature definitions prevent conceptual drift
- Lineage and metadata enable inspection, auditing, and transparency
- Event sequencing, and multi-frequency alignment must be preserved
- Consistent indexing across heterogeneous data
- Temporal coherence is essential for causal and statistical validity

Strategic Replication vs Duplication

- Data lakes support strategic replication to optimize computation without duplication
  - Raw immutable ingest
  - Cleaned/standardized
  - Aggregated/feature-ready
  - Caching/materializing hot subsets

- Retension of authoritative(single) metadata identity and lineage record
  - Meta Data Divergence breaks semantic consistency

Conclusion : Data Lake & LakeHouse

- A data lake or lakehouse is **more than operational infrastructure**; it is a **strategic analytical foundation**.

- It enables **rigorous, reproducible, and scalable** scientific and financial analysis.

- By reducing data entropy and unifying heterogeneous sources, it allows for:
  - **Identify risks earlier and with greater accuracy & transparency**
  - **Discover new opportunities and hidden patterns & transparency**
  - **Accelerate research, decision-making**

- Building such a system is **difficult**: **Not just technology but people & culture is key**
  - Requires **strong design, governance, and stewardship**
  - Without **active maintenance**, **entropy grows and architecture decays**
  - A high-quality lakehouse is **both a capability and a long-term commitment**